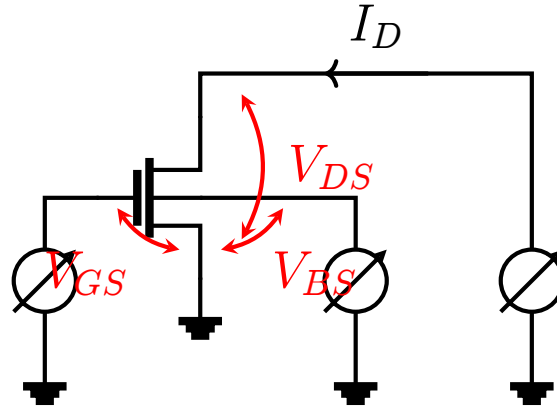


LUT Generation Testbench and idea

Transistor Parameter Formulations & Testbenches



1. Fundamental Transistor Equations

Based on the generic MOS transistor model with parameters Width (W), Length (L), and node voltages (V_{GS} , V_{DS} , V_{BS}):

$$I_D = f_0(W, L, V_{GS}, V_{DS}, V_{BS}) = W \cdot h_0(L, V_{GS}, V_{DS}, V_{BS})$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = f_1(W, L, V_{GS}, V_{DS}, V_{BS}) = W \cdot h_1(L, V_{GS}, V_{DS}, V_{BS})$$

$$g_{ds} = \frac{\partial I_D}{\partial V_{DS}} = f_2(W, L, V_{GS}, V_{DS}, V_{BS}) = W \cdot h_2(L, V_{GS}, V_{DS}, V_{BS})$$

$$C_{gg} = f_3(W, L, V_{GS}, V_{DS}, V_{BS}) = W \cdot h_3(L, V_{GS}, V_{DS}, V_{BS})$$

$$V_{th} = f_4(L, V_{GS}, V_{DS}, V_{BS}) = h_4(L, V_{GS}, V_{DS}, V_{BS})$$

$$g_{mb} = \frac{\partial I_D}{\partial V_{BS}} = f_5(W, L, V_{GS}, V_{BS}) = W \cdot h_5(L, V_{GS}, V_{DS}, V_{BS})$$

2. Figure of Merit Definitions

By taking the ratios of the fundamental parameters, the width (W) cancels out, leaving functions strictly dependent on ($L, V_{GS}, V_{DS}, V_{BS}$), represented by ϕ_n :

$$\left(\frac{g_m}{I_D} \right) = \frac{h_1(L, V_{GS}, V_{DS}, V_{BS})}{h_0(L, V_{GS}, V_{DS}, V_{BS})} = \phi_1(L, V_{GS}, V_{DS}, V_{BS})$$

$$(g_m r_0) = \left(\frac{g_m}{g_{ds}} \right) = \frac{h_1(L, V_{GS}, V_{DS}, V_{BS})}{h_2(L, V_{GS}, V_{DS}, V_{BS})} = \phi_2(L, V_{GS}, V_{DS}, V_{BS})$$

$$f_T = \left(\frac{g_m}{2\pi C_{gg}} \right) = \frac{h_1(L, V_{GS}, V_{DS}, V_{BS})}{2\pi h_3(L, V_{GS}, V_{DS}, V_{BS})} = \phi_3(L, V_{GS}, V_{DS}, V_{BS})$$

Note on Output Resistance & Early Voltage:

$$r_0 = \frac{1}{\lambda I_D} \implies g_{ds} = \lambda I_D \implies \lambda = \left(\frac{g_{ds}}{I_D} \right)$$

$$\lambda = \left(\frac{g_{ds}}{I_D} \right) = \frac{h_2(L, V_{GS}, V_{DS}, V_{BS})}{h_0(L, V_{GS}, V_{DS}, V_{BS})} = \phi_4(L, V_{GS}, V_{DS}, V_{BS})$$

Current Density & Threshold Voltage:

$$\left(\frac{I_D}{W} \right) = h_0(L, V_{GS}, V_{DS}, V_{BS}) \equiv \phi_0(L, V_{GS}, V_{DS}, V_{BS})$$

$$V_{th} = h_4(L, V_{GS}, V_{DS}, V_{BS}) = \phi_5(L, V_{GS}, V_{DS}, V_{BS})$$

Summary of FOMs

- $\left(\frac{g_m}{I_D} \right) = \phi_1(L, V_{GS}, V_{DS}, V_{BS})$
- $(g_m r_0) = \left(\frac{g_m}{g_{ds}} \right) = \phi_2(L, V_{GS}, V_{DS}, V_{BS})$
- $f_T = \left(\frac{g_m}{2\pi C_{gg}} \right) = \phi_3(L, V_{GS}, V_{DS}, V_{BS})$
- $\lambda = \left(\frac{g_{ds}}{I_D} \right) = \phi_4(L, V_{GS}, V_{DS}, V_{BS})$
- $\left(\frac{I_D}{W} \right) = \phi_0(L, V_{GS}, V_{DS}, V_{BS})$
- $V_{th} = \phi_5(L, V_{GS}, V_{DS}, V_{BS})$
- $V^* = \frac{2}{\left(\frac{g_m}{I_D} \right)}$
- $V_A = \frac{1}{\lambda}$

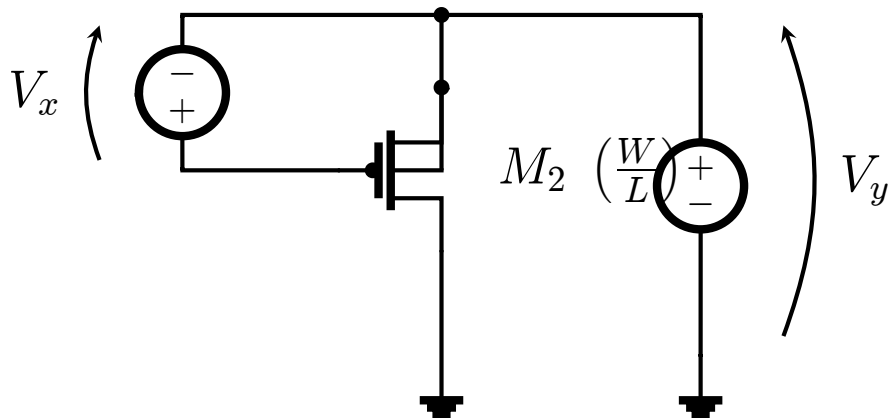
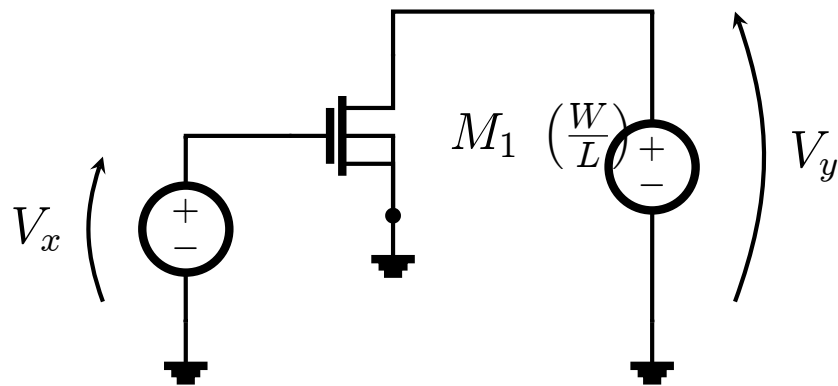
3. Simplified Definitions (Fixed Bias Case)

If we fix the drain-source and bulk-source voltages such that $V_{DS} = \frac{V_{DD}}{2}$ and $V_{BS} = 0$, the parameters become solely dependent on Length (L) and Gate Voltage (V_{GS}):

- $\left(\frac{g_m}{I_D} \right) = \phi_1(L, V_{GS})$
- $(g_m r_0) = \left(\frac{g_m}{g_{ds}} \right) = \phi_2(L, V_{GS})$

- $f_T = \left(\frac{g_m}{2\pi C_{gg}} \right) = \phi_3(L, V_{GS})$
 - $\lambda = \left(\frac{g_{ds}}{I_D} \right) = \phi_4(L, V_{GS})$
 - $\left(\frac{I_D}{W} \right) = \phi_0(L, V_{GS})$
 - $V_{th} = \phi_5(L, V_{GS})$
 - $V^* = \frac{2}{\left(\frac{g_m}{I_D} \right)}$
 - $V_A = \frac{1}{\lambda}$
-

4. Characterization Testbench Setup



Setup Parameters

- $V_y = 1.5 \text{ V}$
- $V_x := 0\text{V} \rightarrow 3\text{V}$ (Step: 0.01V)
- $L := 0.3\mu\text{m} \rightarrow 20\mu\text{m}$ (Step: 10nm)
- $W = 10\mu\text{m}$

Parameters to Save for M1 (NMOS) & M2 (PMOS)

The simulator will run sweeps and store the following parameters as functions of (L, V_x) :

- $I_D(L, V_x)$
 - $g_m(L, V_x)$
 - $g_{mb}(L, V_x)$
 - $g_{ds}(L, V_x)$
 - $V_{th}(L, V_x)$
 - $C_{gs}(L, V_x)$
 - $C_{gd}(L, V_x)$
 - $C_{gb}(L, V_x)$
 - V_{GS} (or V_{SG} for PMOS)
-

5. Parameter Storage Matrix (Look-Up Table Structure)

How the parameters will be stored in data structures:

Any parameter $P(L, V_x)$ will be mapped into an $N \times M$ matrix, where N is the number of length steps and M is the number of voltage steps.

Vector Definitions:

- $L = [L_0, L_1, L_2, \dots, L_{N-1}]^T$ (Size: $N \times 1$)
- $V_x = [V_{x_0}, V_{x_1}, V_{x_2}, \dots, V_{x_{M-1}}]$ (Size: $1 \times M$)

Matrix $P(L, V_x)$:

$$P(L, V_x) = \begin{bmatrix} p(L_0, V_{x_0}) & p(L_0, V_{x_1}) & \dots & p(L_0, V_{x_{M-1}}) \\ p(L_1, V_{x_0}) & p(L_1, V_{x_1}) & \dots & p(L_1, V_{x_{M-1}}) \\ \vdots & \vdots & \ddots & \vdots \\ p(L_{N-1}, V_{x_0}) & p(L_{N-1}, V_{x_1}) & \dots & p(L_{N-1}, V_{x_{M-1}}) \end{bmatrix}$$